Bengaluru House Rent Prediction

https://github.com/aiyshwary/Bengaluru-House-Rent-Prediction

OVERVIEW

A supervised machine learning project that predicts rental prices of houses in Bengaluru using structured feature engineering, multi-stage outlier removal, and benchmarking six regression models with and witho reduction.

IMPACT

Built and benchmarked six regression models on the Bengaluru House Data dataset, finding that Linear F highest accuracy and Decision Tree Regressor performed best without dimensionality reduction, providin pipeline.

KEY DETAILS

1. Loaded and cleaned the Bengaluru_House_Data.csv dataset: dropped irrelevant columns (area_typ removed null rows.

2. Engineered a BHK feature from the size column and normalized the total_sqft column by converting their numeric average.

3. Created a price_per_sqft feature for outlier detection; removed properties with less than 300 sqft per mean±std filtering.

4. Removed BHK outliers by comparing price_per_sqft of each BHK tier against the tier below within th properties where bathrooms exceeded BHK count by more than two.

5. Encoded the location categorical feature using one-hot (dummy) encoding, grouping rare locations v 'other' category to control dimensionality.

6. Benchmarked Decision Tree, Linear Regression, and KNN regressors without PCA; then applied PC
ensembles (90 estimators) for each model.

7. Applied RandomizedSearchCV with 5-fold cross-validation to tune Random Forest Regressor (200 e
max_features, max_leaf_nodes, and min_samples_leaf.

8. Key finding: Linear Regression with PCA achieved the highest overall accuracy; Decision Tree Regr
among non-PCA models.

9. Grouped rare locations appearing 10 or fewer times into an 'other' category, reducing unique locatio
encoding.

10. Removed bathroom outliers where bathrooms exceed BHK count by more than 2; final cleaned dat

11. Visualized 2-BHK vs 3-BHK price scatter plots per location and price-per-sqft histograms with Matp
effectiveness.